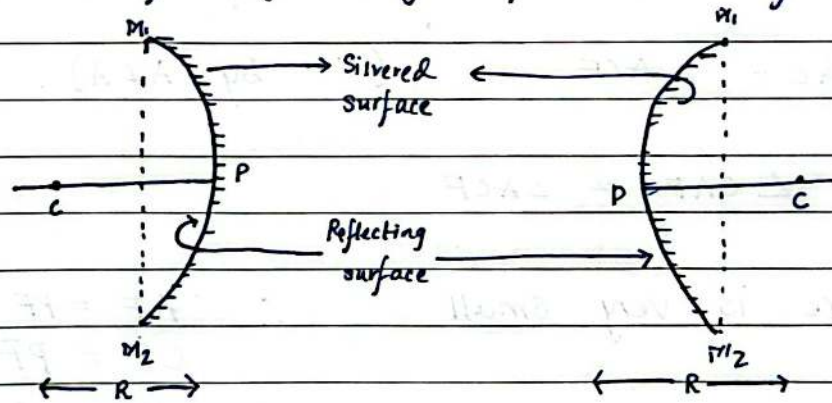


Chapter ~ 9

Ray Optics & Optical Instruments

⇒ Reflection of light by Spherical Surfaces



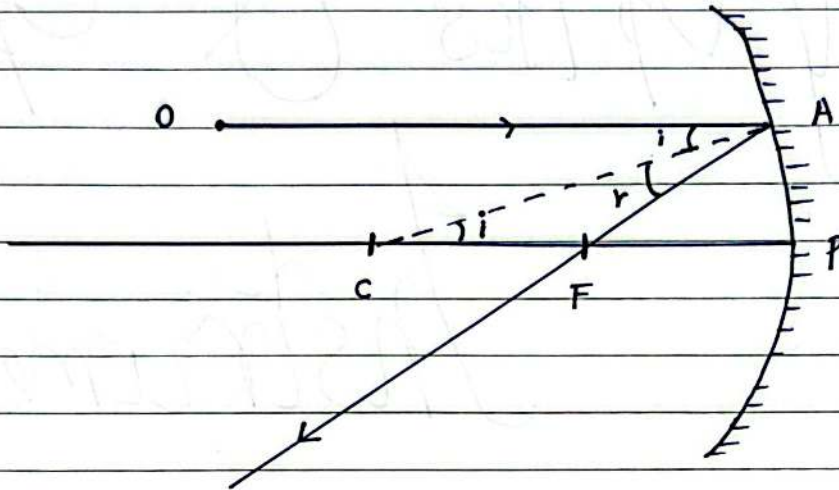
Concave Mirror

Convex mirror

- Sign convention : All distances measured from pole
- * Distances measured in direction of incident light (rightward if it travels left-right) are positive against it are negative.
 - * Heights above principal axis are positive below principal axis are negative

⇒ Relation between focal length (f) and Radius of Curvature (R)

To Prove : $f = R/2$



$$\angle OAC = \angle CAF \quad (\because \text{by } i=r \text{ law}).$$

$$\angle OAC = \angle ACF \quad (\because \text{by } AIA).$$

$$\therefore \angle CAF = \angle ACF$$

Aperture is very small

$$\therefore AF = PF$$

$$CF = PF$$

$$\therefore CF = f$$

$$PC = f + f$$

$$R = 2f$$

$$\boxed{\frac{R}{2} = f}$$

$\Rightarrow$ Position and Nature of image formed by Spherical Mirrors.

Position of object Position of Image Nature
 (A) Concave Mirror

$\Rightarrow$ At Infinity At Focus Real, Inverted
 Highly diminished.

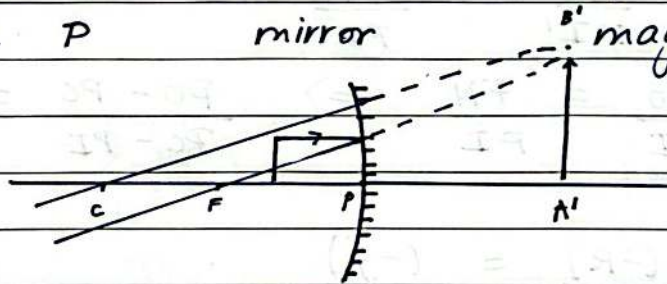
$\Rightarrow$ Beyond C Between C and F Real, Inverted
 Diminished.

$\Rightarrow$ At C At C Real, Inverted
 Same Size.

$\Rightarrow$ Between C and F Between C and ∞ Real, Inverted
 magnified.

$\Rightarrow$ At F At Infinity Real, Inverted
 highly magnified

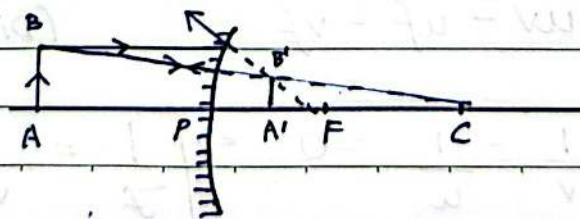
$\Rightarrow$ Between F and P Behind the mirror Virtual Eret
 magnified.



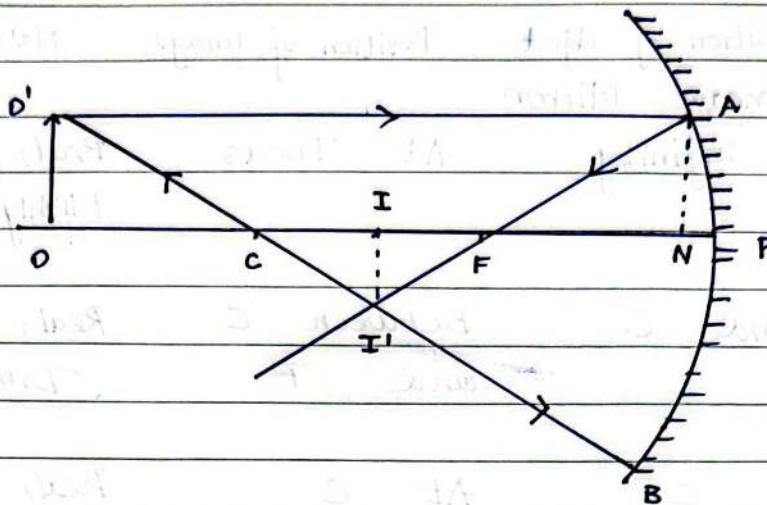
(B) Convex Mirror

$\Rightarrow$ At Infinity At F Virtual, erect
 highly diminished

$\Rightarrow$ At finite distance Between P and F Virtual, erect, diminished.



$\Rightarrow$ Mirror formula for Concave mirror



Take $\triangle OO'C$ & $\triangle II'I'C$

$$\angle C = \angle C \text{ (VOA)}$$

$$\angle O = \angle I \text{ (90°)}$$

$\therefore \triangle OO'C \sim \triangle II'I'C$ (By AA)

$$\frac{OO'}{II'} = \frac{OC}{I'C} = \frac{CO}{CI}$$

Take $\triangle IIF$ & $\triangle NAF$

$$\angle F = \angle F \text{ (VOA)}$$

$$\angle I = \angle N \text{ (90°)}$$

$\therefore \triangle IIF \sim \triangle NAF$ (By AA)

$$\frac{II'}{NA} = \frac{IF}{AF} = \frac{FN}{FM}$$

$$\therefore \frac{OO'}{II'} = \frac{FN}{FI} \quad (NF = PF)$$

$$\text{Now, } \frac{CO}{CI} = \frac{FN}{FI} \Rightarrow \frac{PO - PC}{PC - PI} = \frac{PF}{PI - PF}$$

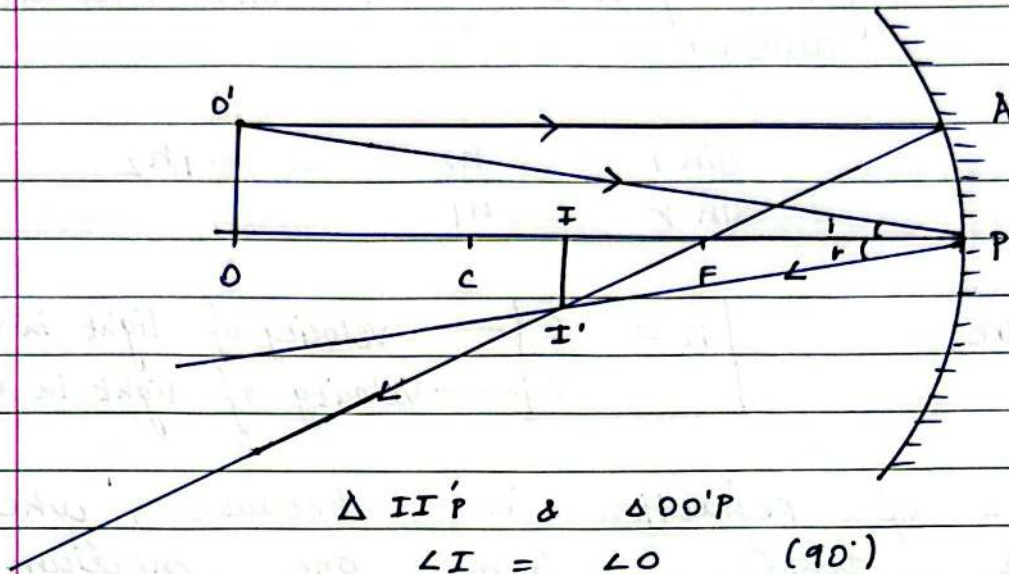
$$\frac{(-u) - (-R)}{(-R) - (-v)} = \frac{(-f)}{(-v) - (-f)}$$

$$\Rightarrow \frac{-u + 2f}{-2f + v} = \frac{-f}{-v + f} \Rightarrow uv - uf - 2fv + 2f^2 = 2f^2 - vf$$

$$uv - uf - vf \quad (\text{Divide } uvf)$$

$$= \frac{1}{f} - \frac{1}{v} - \frac{1}{u} = 0 \Rightarrow \boxed{\frac{1}{f} = \frac{1}{v} + \frac{1}{u}}$$

$\Rightarrow$ Magnification by Spherical Mirrors.



$\Delta II'P$ & $\Delta OO'P$

$$\angle I = \angle O \quad (90^\circ)$$

$$\angle P = \angle P \quad (i = r \text{ By law}).$$

$\Delta II'P \sim \Delta OO'P$ (By AA)

$$\frac{II'}{OO'} = \frac{I'P}{O'P} = \frac{PI}{PO}$$

$$\frac{-I}{+O} = \frac{-v}{-u}$$

$$\boxed{\frac{I}{O} = \frac{-v}{u}}$$

$$\left(\frac{1}{v} + \frac{1}{u} = \frac{1}{f} \right) \times v$$

$$1 - \frac{v}{f} = m$$

$$\boxed{\frac{f-v}{f} = m}$$

$$\left(\frac{1}{v} + \frac{1}{u} = \frac{1}{f} \right) \times u$$

$$1 - \frac{u}{f} = \frac{1}{m}$$

$$\boxed{\frac{f}{f-u} = m}$$

$$\therefore \boxed{m = \frac{-v}{u} = \frac{f-v}{f} = \frac{f}{f-u}}$$

⇒ Reflection of light at a plane interface.

When light passes from one medium to another

$$\frac{\sin i}{\sin r} = \frac{n_2}{n_1} = {}_1n_2$$

Where, $n = \frac{c}{v}$ — velocity of light in vacuum
 — velocity of light in medium.

Cause of refraction is because, when light passes from one medium to another its speed is changed.

⇒ Reversibility of light

When a light ray after suffering any number of reflections or refractions has its final path reversed, it travels back along its initial path.

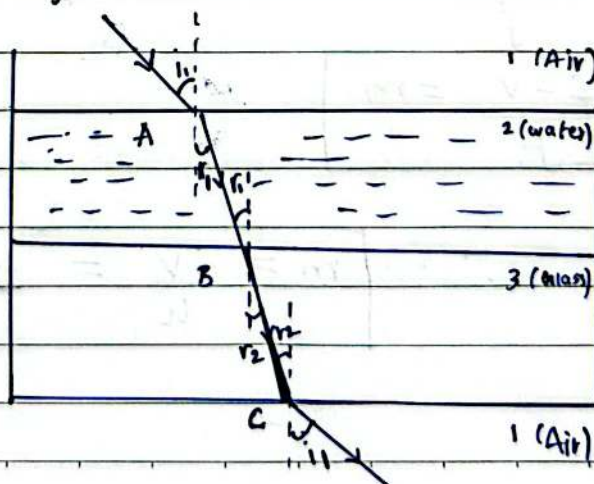
$${}_1n_2 = \frac{1}{{}_2n_1}$$

⇒ Refraction through parallel multiple media.

$${}_1n_2 \times {}_2n_3 = \frac{1}{{}_3n_1}$$

$$\therefore {}_1n_3 = {}_1n_2 \times {}_2n_3$$

$${}_2n_3 = \frac{{}_1n_3}{{}_1n_2}$$



$\Rightarrow$ Real Depth and Apparent Depth

When an object placed in a denser medium is viewed from rarer medium it appears closer to the surface than it actually is because refracted rays bend away from normal on entering rarer medium.

$$a h_w = \frac{R \cdot D}{A \cdot D}$$

$$A \cdot D = \frac{R \cdot D}{a h_w}$$

$$\Rightarrow d = R \cdot D \left[1 - \frac{1}{a h_w} \right]$$

Apparent shift

$$d = R \cdot D - A \cdot D$$

$\Rightarrow$ Total Internal Reflection & Critical Angle.

It occurs when light

- i) It travels from denser to a rarer medium
- ii) Angle of incidence greater than critical angle.

Relation between Refractive Index & Critical Angle

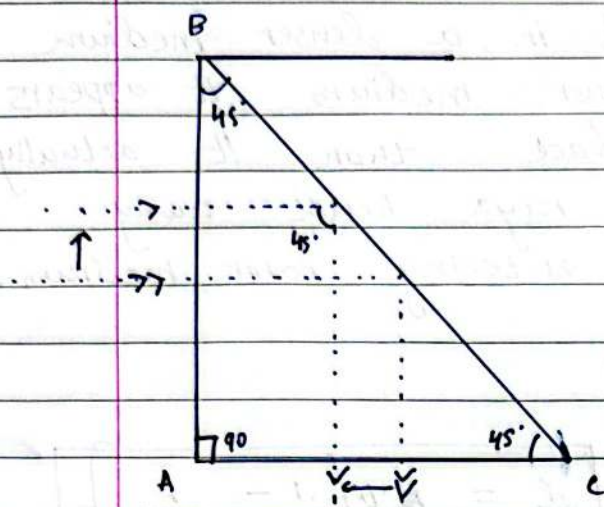
$$n \sin C = 1 \times \sin 90^\circ$$

$$n \sin C = 1$$

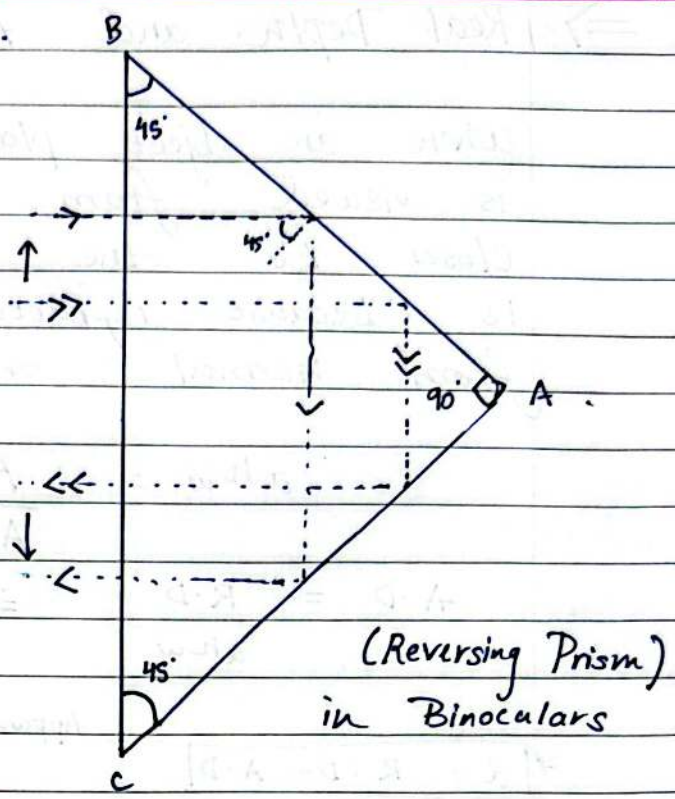
$$\sin C = \frac{1}{n}$$

$\Rightarrow$ Optical Fibres: A thin flexible fibre made of a core of high refractive index glass surrounded by cladding of lower refractive index. It faces multiple TIR along length of fibre over long distances with minimal loss.

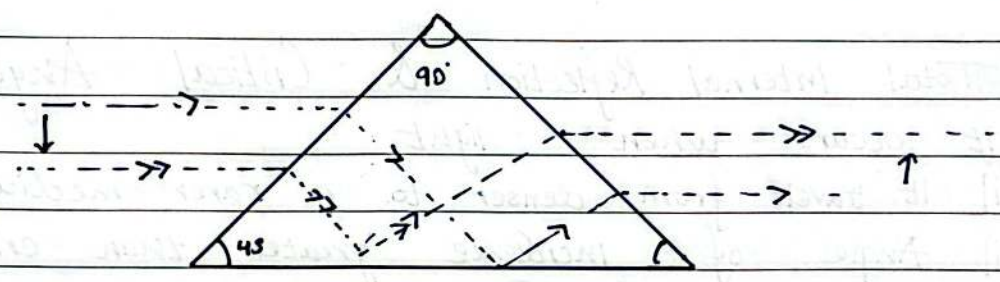
⇒ Total Reflecting Prisms.



(Right Angle Prism)
in periscopes



(Reversing Prism)
in Binoculars

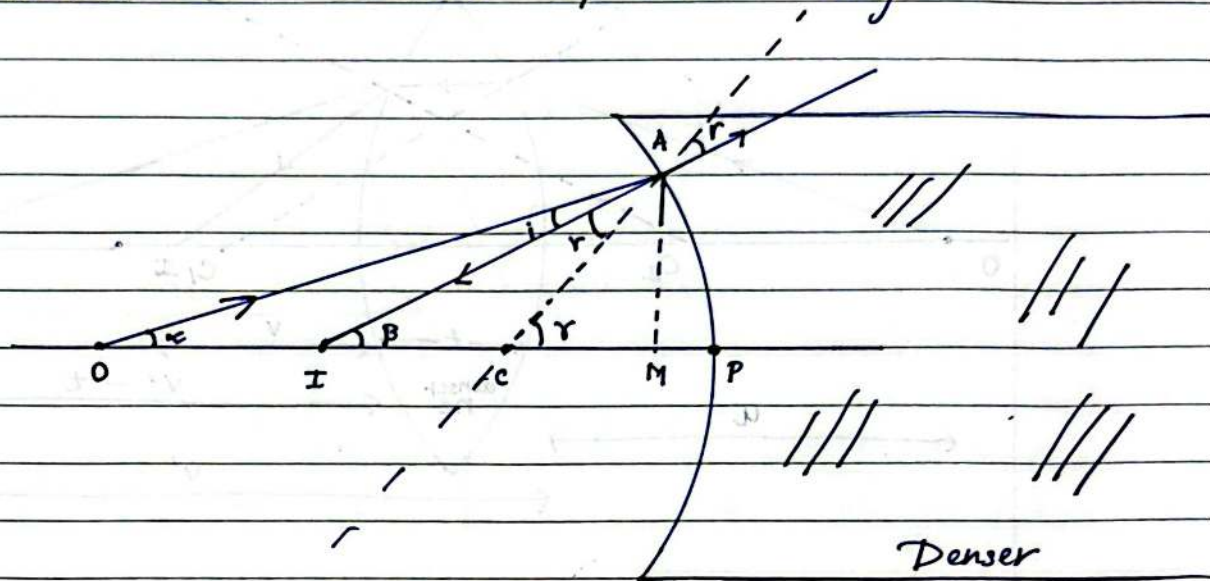


(Erecting Prism)
Prism binoculars.

⇒ Sign convention for u , v and f

⇒ ~~Reflection through a Prism~~, Minimum deviation.
 Refraction of light at Spherical surface.

⇒ Refraction at concave Spherical surface.



$\alpha + i = \gamma$	$B + r = \gamma$	$\tan \alpha = \frac{AM}{OM} \Rightarrow \alpha = \frac{h}{-u}$
$i = \gamma - \alpha$	$r = \gamma - B$	$\tan B = \frac{AM}{IM} \Rightarrow B = \frac{h}{-v}$
$\frac{\sin i}{\sin r} = \frac{n_2}{n_1} \Rightarrow \frac{i}{r} = \frac{n_2}{n_1}$		$\tan \gamma = \frac{AM}{CM} \Rightarrow \gamma = \frac{h}{-R}$

$$(\gamma - \alpha)n_1 = n_2(\gamma - B)$$

$$\left(\frac{h}{-R} - \frac{h}{-u}\right)n_1 = n_2\left(\frac{h}{-R} - \frac{h}{-v}\right)$$

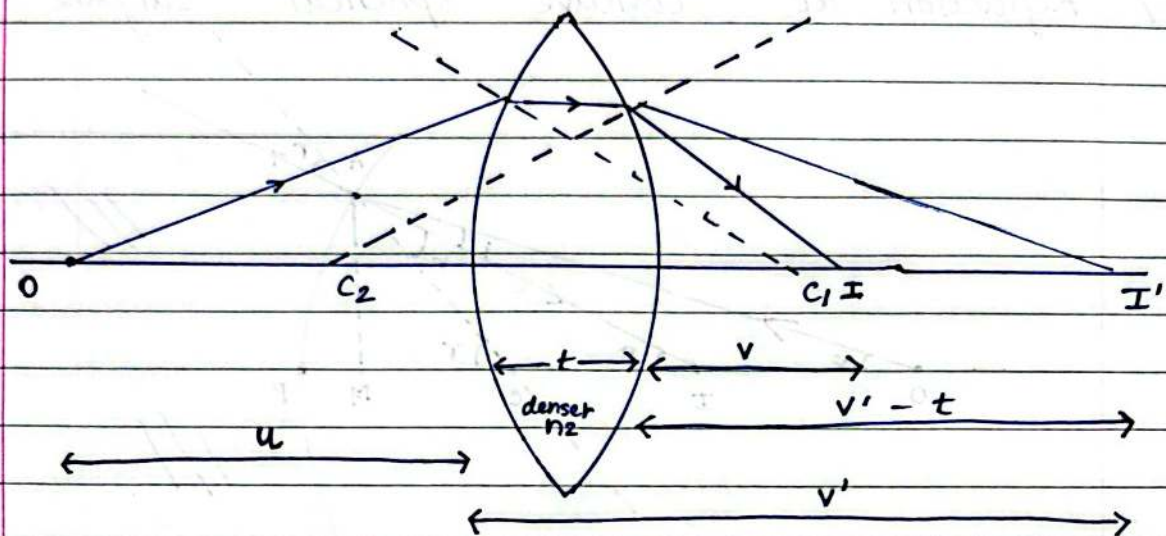
$$\Rightarrow -\frac{n_1}{R} + \frac{n_1}{u} = -\frac{n_2}{R} + \frac{n_2}{v}$$

$$\frac{n_2 - n_1}{R} = \frac{n_2}{v} - \frac{n_1}{u}$$

$$\Rightarrow \boxed{\frac{n_2 - n_1}{R} = \frac{n_2}{v} - \frac{n_1}{u}}$$

$$\boxed{\frac{n_2}{n_1} = \frac{v}{u}}$$

⇒ Refraction of light through a thin lens
 (Lens maker Formula)



For Surface 1

$$\frac{n_2 - n_1}{R_1} = \frac{n_2}{v'} - \frac{n_1}{u}$$

For Surface 2

$$\frac{n_1 - n_2}{R_2} = \frac{n_1}{v} - \frac{n_2}{v' - t}$$

Add Both.

t is very less.

$$\frac{n_2 - n_1}{R_1} + \frac{n_1 - n_2}{R_2} = \frac{n_1}{v} - \frac{n_2}{u}$$

$$(n_2 - n_1) \left[\frac{1}{R_1} - \frac{1}{R_2} \right] = \frac{n_1}{v} - \frac{n_2}{u}$$

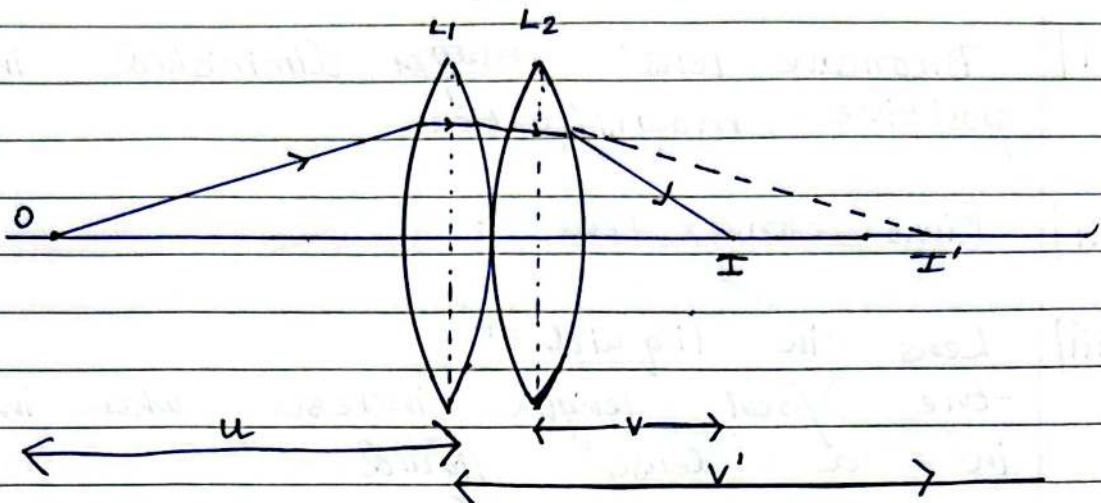
$$(n - 1) \left[\frac{1}{R_1} - \frac{1}{R_2} \right] = \frac{1}{v} - \frac{1}{u}$$

⇒ Special Case : If $u = \infty$ then $v = f$

$$(n - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right) = \frac{1}{f}$$

⇒ Thin lens formula ⇒ $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$

⇒ Combined Focal length of two thin lens



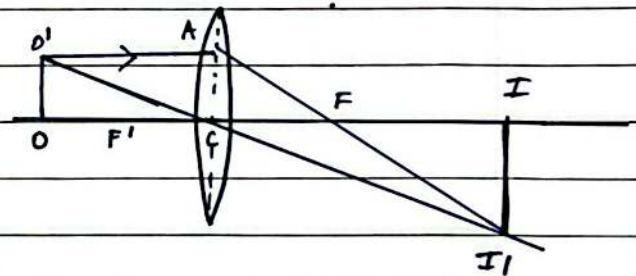
$$L_1 = \frac{1}{v} - \frac{1}{u} = \frac{1}{f_1} \quad \bigg| \quad L_2 = \frac{1}{v'} - \frac{1}{v} = \frac{1}{f_2}$$

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f_1} + \frac{1}{f_2} \quad \text{or } P = P_1 + P_2.$$

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f} \Rightarrow \boxed{\frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2}}$$

⇒ Magnification by a biconvex lens

$\Delta CII'$ and $\Delta COO'$
 are similar.



$$\frac{II'}{OO'} = \frac{CI}{CO}$$

$$\frac{-h_2}{h_1} = \frac{v}{-u}$$

$$\Rightarrow \boxed{\frac{h_2}{h_1} = \frac{v}{u} = m}$$

⇒ Types of lens.

i) Biconcave lens: ~~magn~~ diminished image.
 positive magnification

ii) Plano convex lens. :

iii) Lens in liquid :
 the focal length increases when immersed
 in a denser liquid.

⇒ Power of a lens.

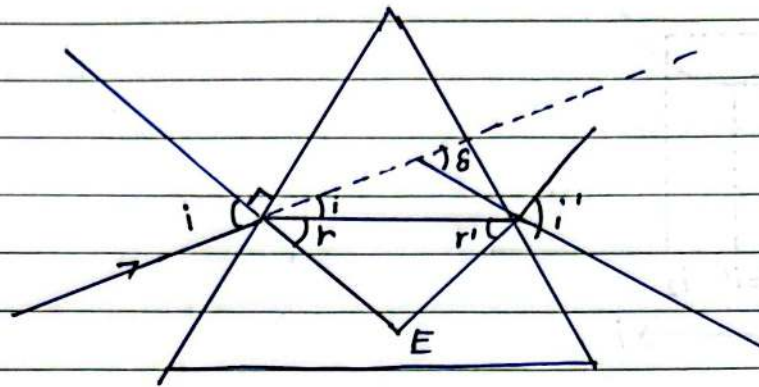
$$P = \frac{1}{f}$$

unit = Dioptre

convex = +ve power

concave = -ve power.

$\Rightarrow$ Refraction through a Prism.



When $\delta = \delta_m \Rightarrow i' = i$ & $r' = r$

$$A + E + 90 + 90 = 360$$

$$A + E = 180$$

$$A = 180 - E$$

$$r + r' + E = 180$$

$$r + r' = 180 - E$$

$$\therefore \boxed{A = r + r'} \quad A = 2r$$

$$\text{Now, } (i - r) + (i' - r') = \delta$$

$$(i + i') - (r + r') = \delta$$

$$2i - 2r = \delta_{\min}$$

$$\boxed{2i - A = \delta_{\min}}$$

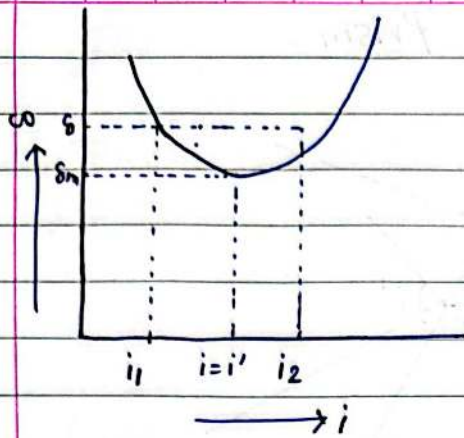
$$\text{Now, } n = \frac{\sin i}{\sin r}$$

$$n = \frac{\sin \left(\frac{A + \delta_m}{2} \right)}{\sin (A/2)}$$

Special Case: thin prism $n = \frac{A + \delta_m}{2}$

$$\Rightarrow nA = A + \delta_m$$

$$A(n - 1) = \delta_m$$



⇒ Dispersion: Splitting of white light into its seven constituent colours.

⇒ Angular Dispersion: Difference between two extreme colours

$$\delta_v - \delta_r = (n_v - n_r)A$$

⇒ Dispersive power (ω)
 How strongly materials spreads out its colours

$$\omega = \frac{n_v - n_r}{n_y - 1}$$

⇒ Rainbow.

Its formed due to dispersion, refraction, total internal reflection.

Primary Rainbow

1. one TIR and two refractions
2. It is brighter

Secondary Rainbow.

- two TIR and two refractions
- It is fainter.